

PrepAIred: A Reinforcement-Based LLM-Powered Technical Interview Preparation System

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Abstract—Technical interview preparation requires candidates preparing for software engineering roles to demonstrate technical knowledge, algorithmic reasoning, problem-solving ability, explanation clarity, and verbal communication under realistic evaluation conditions. Achieving interview readiness is essential for computer science students and job seekers navigating competitive software hiring processes, yet human mock interviews present severe limitations in accessibility, scalability, financial cost, and evaluator variability. Conversely, existing automated evaluation systems exhibit critical deficiencies: static question repositories apply fixed difficulty schedules regardless of individual capability, single-score evaluation metrics fail to isolate technical understanding from speech delivery, semantic similarity measures are easily fooled by keyword repetition without logical structure, and unconstrained Large Language Model interviewers suffer from prompt drift, hallucinated grading criteria, and an inability to provide grounded follow-up questioning. To resolve these limitations, there is an urgent design gap for an integrated interview-preparation framework capable of evaluating technical explanations across multiple structured signals, analyzing speech communication independently, adapting question difficulty dynamically, identifying unresolved conceptual gaps, asking grounded follow-up questions, and providing qualitative candidate feedback within a single continuous interview session. This paper presents PrepAIred, an adaptive multimodal technical interview preparation system that decouples quantitative evaluation from qualitative conversational generation. PrepAIred evaluates candidate responses using a hybrid technical evaluator combining SentenceTransformer semantic similarity, rubric-based concept coverage, and fine-tuned CrossEncoder reasoning verification, operating alongside an independent speech confidence model that extracts acoustic, fluency, and temporal delivery features. Question difficulty adaptation is governed by a Proximal Policy Optimization reinforcement learning agent observing a continuous six-dimensional candidate state representation, while a deterministic decision engine gates Qwen2.5-7B-Instruct served through local Ollama to generate grounded follow-up probes and qualitative candidate feedback without exposing internal numerical scores. System verification demonstrates strong reasoning alignment with a CrossEncoder Spearman rank correlation of 0.796 on evaluation splits, while an implementation ablation confirms that eliminating redundant LLM calls reduces session evaluation latency from 73.22 seconds to 34.01 seconds and memory pressure from 14.17 gigabytes to 11.20 gigabytes without score drift.

Index Terms—Adaptive Tutoring Systems, Natural Language Processing, Reinforcement Learning, Large Language Models, Multimodal Assessment, Technical Interview Evaluation.

I. INTRODUCTION

Technical interviews in computer science, data structures, algorithms, and system design require candidates to simultaneously demonstrate domain knowledge, analytical problem-solving, structured logic, and fluent communication. For students and job seekers, mock interview practice is an essential component of career preparation. However, scalable mock interview platforms present fundamental trade-offs. Human mock interviews provide high-quality, contextual feedback but are costly, non-scalable, and subject to evaluator variability. Conversely, automated preparation systems traditionally rely on static question repositories with pre-scripted test cases, failing to adapt dynamically when a candidate exhibits partial conceptual understanding, misunderstands a core requirement, or demonstrates strong technical correctness alongside hesitant verbal delivery.

The recent emergence of general-purpose Large Language Models (LLMs) has enabled conversational interview agents capable of generating dynamic questions and natural language feedback. Despite their fluency, unconstrained LLM interviewers introduce significant operational risks in educational assessment. Unbounded LLM evaluators often suffer from grading inconsistency, prompt drift, hallucinated rubric criteria, and an inability to maintain rigid numerical scoring standards. Furthermore, unconstrained LLM prompts frequently conflate polished verbal articulation with technical correctness, allowing candidates to achieve favorable evaluations through keyword repetition or superficial phrasing even when core reasoning is flawed. Crucially, standalone LLMs lack principled difficulty adaptation mechanisms, relying instead on subjective prompt interpretations that fail to enforce structured pedagogical progression.

To address these deficiencies, PrepAIred introduces a decoupled multimodal architecture that separates rigorous quantitative technical and speech assessment from qualitative natural language generation. PrepAIred orchestrates technical answer evaluation, speech delivery analysis, reinforcement-learning-driven difficulty control, evidence-gated follow-up generation, and grounded qualitative feedback into a unified, continuous interview loop.

The operational pipeline integrates five core functional mechanisms. First, a hybrid technical answer evaluator combines SentenceTransformer semantic similarity, rubric-based structural concept coverage, and CrossEncoder reasoning entailment, supplemented by explicit mistake penalties, bonus credits, and mandatory concept constraints. Second, an independent speech communication confidence model processes prosodic quality, pause dynamics, speaking rate, rhythm, and speech recognition confidence into a distinct delivery signal maintained separately from technical content accuracy. Third, an adaptive difficulty controller powered by Proximal Policy Optimization (PPO) observes a continuous six-dimensional candidate state vector to select optimal difficulty adjustments across structured question pools. Fourth, a deterministic follow-up state machine evaluates structured evaluator evidence to trigger targeted follow-up probing using Qwen2.5-7B-Instruct served locally via Ollama, bounding follow-up attempts to prevent endless conversational drift. Fifth, a candidate-facing feedback engine generates comprehensive qualitative guidance summarizing covered concepts, identified misconceptions, and delivery feedback while strictly suppressing internal numerical scores to foster a low-anxiety learning environment.

The main contributions of this work are fourfold: a multi-tier technical evaluation pipeline unifying semantic, structural, and pairwise reasoning verification; an independent acoustic and temporal communication model integrated directly into reinforcement learning state space; an evidence-driven state machine that gates LLM generation based on deterministic evaluation triggers; and an empirical evaluation confirming system optimization and functional verification across complex interview scenarios.

II. RELATED WORK

Automated short-answer scoring and technical assessment have evolved from early surface-level keyword matching to modern transformer-based semantic representations. SentenceBERT demonstrated that Siamese transformer networks can produce semantically meaningful sentence embeddings for fast similarity comparisons [1]. However, semantic similarity alone remains vulnerable to false positives when candidate responses contain relevant terminology without correct logical structure. To evaluate pairwise reasoning context, CrossEncoder models trained on passage ranking datasets such as MS MARCO provide fine-grained entailment verification [2]. While previous automated assessment research focuses primarily on static answer grading, PrepAIred integrates CrossEncoder reasoning verification into a broader adaptive interview system that connects quantitative scoring directly to downstream question selection and grounded follow-up generation.

Speech analysis and communication modeling in professional interviews have established that non-verbal prosodic features strongly correlate with perceived confidence and interview readiness. Classical acoustic toolkits such as Praat enable the extraction of physical voice parameters including jitter, shimmer, and Harmonics-to-Noise Ratio (HNR) [3].

Subsequent interview prediction systems utilized multimodal feature extraction, combining doc2vec language representations with acoustic speech parameters to predict human ratings [22], while automated interview video frameworks applied multimodal regression to evaluate candidate delivery [21]. However, existing speech-enabled interview tools frequently combine acoustic metrics into an overall performance score, potentially penalizing technically proficient candidates who exhibit vocal hesitation. PrepAIred differs by maintaining speech confidence as an independent, non-penalizing state dimension that informs difficulty selection without degrading technical correctness scores.

Reinforcement Learning (RL) and Intelligent Tutoring Systems (ITS) have long addressed adaptive pedagogical sequencing. Early cognitive tutors modeled student skill acquisition using Bayesian Knowledge Tracing [18] and rule-based cognitive architectures [17]. Modern curriculum reinforcement learning formulates educational path optimization as a Markov Decision Process (MDP), utilizing policy gradient algorithms such as Proximal Policy Optimization (PPO) to dynamically balance challenge and candidate capability [4], [19], [20]. Adaptive testing frameworks grounded in item response theory similarly seek to optimize test length and question difficulty [16]. While traditional RL tutoring models adjust content across structured static lessons, PrepAIred applies PPO difficulty adaptation within a multi-turn, speech-interactive technical interview environment governed by a continuous six-dimensional observation state.

Large Language Models (LLMs) in educational assessment offer flexible narrative feedback but suffer from unconstrained generation and hallucination risks [6], [8]. Modern instructional frameworks utilize retrieval-augmented generation and prompt grounding to restrict LLM outputs within defined boundary conditions. In technical interview simulation, unconstrained LLM interviewers frequently drift from rubric definitions or fail to enforce consistent difficulty escalation. PrepAIred resolves this limitation by stripping the LLM of scoring authority, utilizing Qwen2.5-7B-Instruct exclusively through local Ollama inference as a grounded natural language generator operating strictly under deterministic evidence-driven state machine triggers.

Multimodal interview simulation frameworks have recently been extended to address broader career guidance problems. The Ascend AI framework provides an intelligent multimodal architecture for personalized career direction and adaptive technical interview simulation [23]. Ascend AI addresses the end-to-end career guidance pipeline by unifying psychometric candidate profiling, predictive career-path modeling, personalized learning-roadmap generation, and interview simulation. PrepAIred differs in scope and architectural focus from Ascend AI: while Ascend AI addresses broader career guidance, recommendation, and learning-roadmap generation, PrepAIred focuses specifically on single-session adaptive technical interview assessment and difficulty optimization. PrepAIred addresses the technical evaluation gap by combining question-specific hybrid answer evaluation, an independent acoustic

TABLE I
COMPARATIVE ANALYSIS OF RELATED LITERATURE AND PREPAIRED

Research Theme	Representative Works	Scope / Limitations	PrepAired Approach
Automated Answer Assessment	Reimers & Gurevych [1], Bajaj et al. [2]	Static scoring without interactive questioning	Combines SBERT, rubric coverage, and CrossEncoder into adaptive loop
Speech & Communication	Boersma [3], Naim et al. [21], Chen et al. [22]	Conflates speech metrics into overall score	Separates speech confidence as an independent RL state dimension
Adaptive Tutoring & RL	Anderson et al. [17], Schulman et al. [4], Graves et al. [19]	Applied to fixed static or multiple-choice tasks	Applies PPO over 6D state targeting technical interviews
LLMs in Assessment	Ji et al. [6], Qwen Team [8]	Unconstrained scoring causes rubric drift	Restricts LLM to grounded follow-ups gated deterministically
Multimodal Career Frameworks	Ascend AI [23]	Focuses on career paths and learning roadmaps	Focuses specifically on technical answer evaluation and 6D PPO adaptation

communication confidence model, PPO difficulty adaptation over a continuous six-dimensional state, evidence-gated deterministic follow-up selection, bounded Qwen2.5-7B follow-up probing, and qualitative feedback within single-column execution constraints.

Table I synthesizes the key domain literature, contrasting existing approaches against the integrated methodology implemented in PrepAired within single-column boundaries.

III. SYSTEM ARCHITECTURE

The PrepAired system is built on a decoupled microservice architecture designed to ensure low-latency WebSocket communication, deterministic grading, and resilient local LLM serving. The system comprises a React 18 single-page application, a FastAPI and WebSocket session gateway, an asynchronous orchestration engine, a PyTorch technical evaluation service, an audio feature processing module, a Stable-Baselines3 PPO difficulty controller, and a local Ollama service running Qwen2.5-7B-Instruct.

The end-to-end operational lifecycle of PrepAired is illustrated in Figure 1. The workflow executes within strict single-column constraints, representing the complete session lifecycle from initial configuration to final session feedback.

The session begins when a candidate accesses the single-page application interface and selects a session length configuration of 5, 15, or 30 primary questions. Upon session initialization, the system samples Question 1 randomly from the Easy question pool. When the candidate presents their verbal or written response, the input stream is captured and processed through parallel analysis pipelines. Spoken responses are transcribed using Automatic Speech Recognition (ASR) while raw audio signals are dispatched to the prosodic feature extraction engine. The resulting transcript T_c is passed to the hybrid technical evaluator while acoustic features feed the communication confidence model.

The evaluator extracts three core technical scores: semantic similarity S_1 , structural concept coverage S_2 , and CrossEncoder reasoning score R . In parallel, the communication model calculates composite delivery confidence C_{final} . These

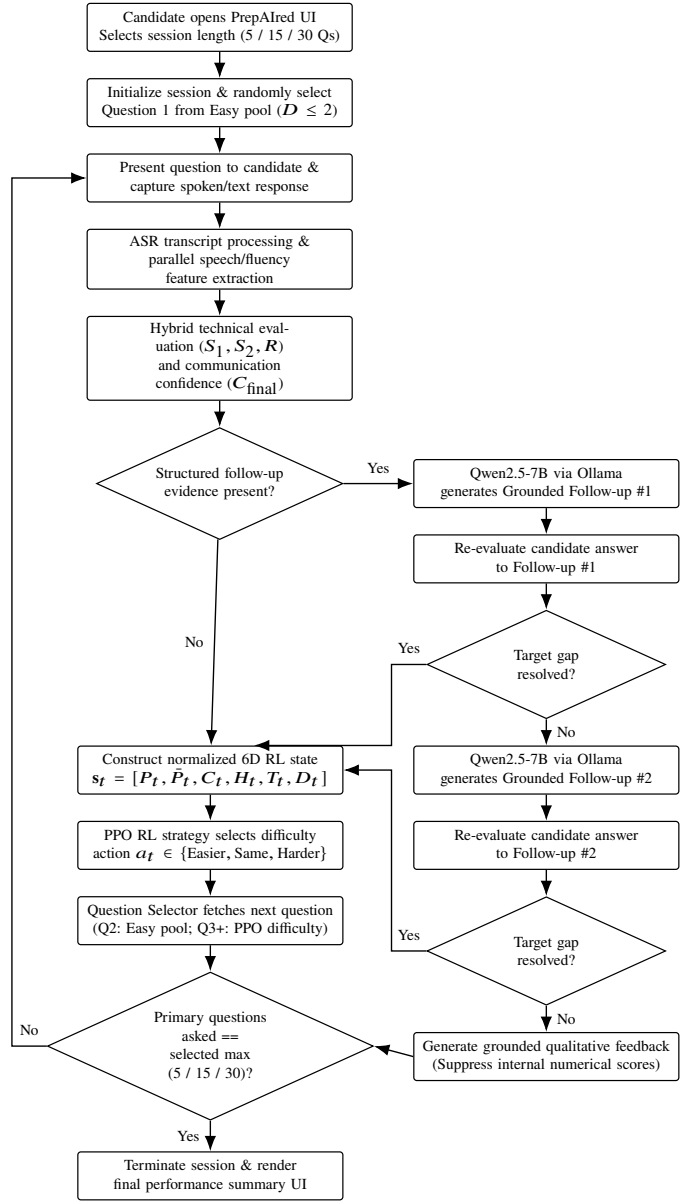


Fig. 1. End-to-end operational workflow of PrepAired within a single-column layout. The candidate configures session parameters, establishing a two-question baseline phase. Text transcripts and speech audio undergo parallel evaluation to yield technical correctness and communication confidence scores. Deterministic evidence logic determines whether to initiate grounded Qwen follow-up probing (up to 2 bounded attempts) or proceed directly through the 6D PPO state adaptation loop until the target primary question count is fulfilled.

outputs feed into a deterministic follow-up decision layer. If structured evidence indicates an unresolved conceptual gap or misconception, the system diverts from normal question progression to launch a grounded follow-up sequence powered by Qwen2.5-7B-Instruct. The follow-up sub-loop permits up to two targeted attempts. If the candidate resolves the gap, control returns to the primary adaptive progression; if the gap remains unresolved after two attempts, the system generates comprehensive qualitative feedback and proceeds to the next primary turn.

TABLE II
PREPAIRED ARCHITECTURAL COMPONENT SPECIFICATION

Component	Method / Library	Operational Role
Web Frontend	React 18 SPA + Web Speech API	User interaction, audio capture, score-suppressed UI
API Gateway	FastAPI + WebSockets	Real-time session routing and state orchestration
Hybrid Evaluator	SentenceTransformers + FAISS + CrossEncoder	Deterministic technical answer scoring
Confidence Engine	Praat acoustic features + timing analysis	Speech delivery and vocal confidence modeling
RL Strategy	Stable-Baselines3 PPO	Continuous state difficulty adaptation
LLM Engine	Qwen2.5-7B-Instruct via Ollama	Grounded follow-up question and feedback generation
Question Bank	Structured JSON / FAISS Indices	Rubrics, concept vectors, and question pools

For turns that do not require follow-up probing, or upon completion of a follow-up sub-loop, the orchestrator constructs a continuous six-dimensional observation vector $\mathbf{s}_t$. The PPO RL strategy evaluates $\mathbf{s}_t$ and selects a discrete difficulty adjustment action $a_t \in \{\text{Easier, Same, Harder}\}$. The question selector retrieves a matching question from the target difficulty tier, enforcing an intra-session non-repetition constraint. This adaptive cycle repeats until the total number of primary questions asked equals the candidate’s initial session length selection, at which point the session terminates and renders the final performance feedback view.

Table II details the primary technical modules, underlying models, and designated roles within the runtime architecture.

IV. HYBRID TECHNICAL ANSWER EVALUATION

The technical answer evaluation engine calculates a quantitative score $S_{\text{final}} \in [0.0, 1.0]$ by evaluating candidate transcript T_c against canonical question rubric R_q . Rubric R_q contains semantic reference statements R_q^{semantic} , required technical concepts R_q^{concept} , common misconception vectors M_q , advanced insight vectors B_q , and mandatory concept flags.

A. Semantic Similarity (S_1)

Candidate transcript T_c is segmented into n constituent sentences $S = \{s_1, s_2, \dots, s_n\}$. Each sentence s_i and semantic reference statement $t_j \in R_q^{\text{semantic}}$ are mapped to 384-dimensional dense vector embeddings using the all-MiniLM-L6-v2 SentenceTransformer model, yielding candidate sentence vectors $\mathbf{e}_i = \text{Embed}(s_i)$ and reference vectors $\mathbf{v}_j = \text{Embed}(t_j)$. The semantic similarity score S_1 is computed as the mean of the maximum cosine similarities between each rubric reference and the candidate sentence embeddings:

$$S_1 = \frac{1}{|R_q^{\text{semantic}}|} \sum_{j=1}^{|R_q^{\text{semantic}}|} \max_{1 \leq i \leq n} \cos(\mathbf{e}_i, \mathbf{v}_j), \quad (1)$$

where $|R_q^{\text{semantic}}|$ denotes the total number of semantic reference statements in the rubric, and $\cos(\mathbf{u}, \mathbf{w}) = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\| \|\mathbf{w}\|}$ represents cosine similarity.

B. Concept Coverage (S_2) and Dampening

Structural concept coverage S_2 evaluates the inclusion of required domain concepts R_q^{concept} . For each concept vector $\mathbf{c}_k = \text{Embed}(c_k)$ associated with concept $c_k \in R_q^{\text{concept}}$, the system determines concept presence using indicator function $\mathbb{I}(\cdot)$ subject to concept similarity threshold $\tau_{\text{concept}} = 0.30$:

$$S_2 = \frac{1}{|R_q^{\text{concept}}|} \sum_{k=1}^{|R_q^{\text{concept}}|} \mathbb{I} \left(\max_{1 \leq i \leq n} \cos(\mathbf{e}_i, \mathbf{c}_k) > 0.30 \right), \quad (2)$$

where $|R_q^{\text{concept}}|$ is the number of required concepts. To eliminate vulnerability to keyword stuffing wherein candidates list domain terms without explaining logical relationships, effective concept coverage $S_2^{\text{effective}}$ is dampened when reasoning score R falls below minimum reasoning floor 0.30:

$$S_2^{\text{effective}} = \begin{cases} S_2, & R > 0.30, \\ 0.60 \cdot S_2, & R \leq 0.30. \end{cases} \quad (3)$$

This dampening mechanism ensures that candidate answers mentioning relevant terms without establishing reasoning context receive penalized concept coverage credit.

C. CrossEncoder Reasoning Verification (R)

Reasoning entailment R evaluates whether the candidate transcript correctly substantiates the technical claim relative to expected reference answer A_q . Candidate transcript T_c and reference A_q are concatenated and processed through cross-encoder/ms-marco-MiniLM-L-6-v2. The raw CrossEncoder logit output r_{raw} is transformed into normalized score $R \in [0.0, 1.0]$ via linear scaling:

$$R = \text{clamp} \left(\frac{r_{\text{raw}} - 0.20}{0.70}, 0.0, 1.0 \right), \quad (4)$$

where $\text{clamp}(x, a, b) = \min(b, \max(a, x))$ restricts values to $[a, b]$. Baseline logit threshold 0.20 defines minimum reasoning alignment, while scaling factor 0.70 maps raw logits across the unit interval.

D. Bonus, Mistake Penalty, and Mandatory Concepts

Candidate sentences are scanned against mistake vectors M_q and bonus insight vectors B_q . Total mistake penalty P_{mistake} accumulates matched penalty increments subject to penalty cap 0.30:

$$P_{\text{mistake}} = \min \left(0.30, \sum_{m \in M_q} 0.07 \cdot \mathbb{I} \left(\max_{1 \leq i \leq n} \cos(\mathbf{e}_i, \mathbf{m}) > 0.55 \right) + P_{\text{reasoning}} \right), \quad (5)$$

where $\mathbf{m}$ is a mistake embedding vector, 0.55 represents mistake matching threshold, 0.07 is per-mistake penalty weight, and $P_{\text{reasoning}}$ represents additional penalties assigned for structural reasoning failures.

TABLE III
HYBRID TECHNICAL EVALUATOR PARAMETER CONFIGURATIONS

Parameter Name	Value	Functional Description
Semantic Weight (S_1)	0.15	Weight assigned to SBERT similarity
Concept Weight (S_2)	0.35	Weight assigned to structural concept coverage
Reasoning Weight (R)	0.50	Weight assigned to CrossEncoder reasoning
Concept Threshold (τ_{concept})	0.30	Cosine similarity cutoff for concept match
Reasoning Floor	0.30	CrossEncoder threshold triggering dampening
Dampening Factor	0.60	Concept coverage multiplier when $R \leq 0.30$
Bonus Increment	0.03	Score credit per matched advanced insight
Bonus Threshold	0.50	Cosine similarity cutoff for bonus match
Mistake Increment	0.07	Score penalty per matched mistake vector
Mistake Threshold	0.55	Cosine similarity cutoff for mistake match
Penalty Cap	0.30	Maximum accumulated mistake penalty
Mandatory Cap	0.60	Maximum score if mandatory concepts missed

Bonus credit S_{bonus} rewards advanced insights:

$$S_{\text{bonus}} = \sum_{b \in B_q} 0.03 \cdot \mathbb{I} \left(\max_{1 \leq i \leq n} \cos(\mathbf{e}_i, \mathbf{b}) > 0.50 \right), \quad (6)$$

where $\mathbf{b}$ is a bonus insight embedding vector, 0.50 is bonus matching threshold, and 0.03 is bonus credit weight.

The base technical score S_{base} aggregates weighted components:

$$S_{\text{base}} = 0.15 \cdot S_1 + 0.35 \cdot S_2^{\text{effective}} + 0.50 \cdot R. \quad (7)$$

The uncapped preliminary score is adjusted for bonuses and penalties:

$$S_{\text{final}} = \text{clamp}(S_{\text{base}} + S_{\text{bonus}} - P_{\text{mistake}}, 0.0, 1.0). \quad (8)$$

If the rubric designates mandatory concepts and the candidate fails to cover all mandatory concepts, final score S_{final} is constrained by hard mandatory score cap 0.60:

$$\text{Mandatory Pass} = \text{False} \implies S_{\text{final}} = \min(0.60, S_{\text{final}}). \quad (9)$$

Table III summarizes the verified technical parameters utilized by the evaluation engine.

V. COMMUNICATION CONFIDENCE MODEL

Candidate vocal delivery and speech communication are modeled independently of technical correctness. This structural decoupling ensures that vocal hesitation or delivery nervousness does not alter technical answer scores, while allowing the RL difficulty policy to adapt to candidate confidence state.

A. Prosodic Voice Quality (V_Q)

Prosodic analysis extracts fundamental acoustic parameters: pitch jitter j , pitch shimmer s , Harmonics-to-Noise Ratio h , and pitch variance $C_{\text{pitch_var}}$. Normalized acoustic metrics are computed using empirical scale parameters:

$$C_{\text{jitter}} = \text{clamp}(1.0 - j/0.03, 0.0, 1.0), \quad (10)$$

$$C_{\text{shimmer}} = \text{clamp}(1.0 - s/0.12, 0.0, 1.0), \quad (11)$$

$$C_{\text{HNR}} = \text{clamp}(h/28.0, 0.0, 1.0), \quad (12)$$

where 0.03 represents maximum expected jitter, 0.12 represents maximum expected shimmer, and 28.0 dB represents reference high-quality HNR. Composite voice quality V_Q combines normalized prosodic signals:

$$V_Q = 0.32 \cdot C_{\text{jitter}} + 0.30 \cdot C_{\text{shimmer}} + 0.28 \cdot C_{\text{HNR}} + 0.10 \cdot C_{\text{pitch_var}}. \quad (13)$$

B. Speech Fluency, Rate, and Rhythm

Temporal speech characteristics are evaluated using pause ratio, Words Per Minute (WPM), and pause counts. Speaking fluency F_{fluency} penalizes excessive silent pauses relative to total speech duration:

$$F_{\text{fluency}} = \text{clamp} \left(1.0 - 1.7 \cdot \frac{t_{\text{pause}}}{t_{\text{speech}}}, 0.0, 1.0 \right), \quad (14)$$

where t_{pause} is total detected pause time, t_{speech} is active speech duration, and 1.7 is empirical pause scaling coefficient.

Speaking rate fluency F_{rate} measures pacing relative to optimal interview speech rate bounds [2.1, 4.3] WPM:

$$F_{\text{rate}} = \begin{cases} 1.0, & 2.1 \leq \text{WPM} \leq 4.3, \\ \text{clamp} \left(1.0 - \frac{|\text{WPM} - 3.2|}{3.2}, 0.0, 1.0 \right), & \text{otherwise,} \end{cases} \quad (15)$$

where 3.2 WPM represents target center speech rate. Rhythm score F_{rhythm} penalizes frequent erratic pauses beyond threshold $N_p = 7$:

$$F_{\text{rhythm}} = \text{clamp} \left(1.0 - \frac{\max(0, N_p - 7)}{12}, 0.0, 1.0 \right), \quad (16)$$

where N_p is total pause count and 12 is rhythm scaling factor.

C. Composite Communication Confidence (C_{final})

Composite communication confidence C_{final} aggregates vocal quality, linguistic confidence L , fluency, speaking rate, and rhythm:

$$C_{\text{final}} = 0.28 \cdot V_Q + 0.27 \cdot L + 0.22 \cdot F_{\text{fluency}} + 0.13 \cdot F_{\text{rate}} + 0.10 \cdot F_{\text{rhythm}}. \quad (17)$$

If ASR transcript confidence A_{conf} drops below quality threshold 0.45, the model adjusts reliance toward prosodic features:

$$A_{\text{conf}} < 0.45 \implies C_{\text{final}} = 0.75 \cdot C_{\text{final}} + 0.25 \cdot V_Q. \quad (18)$$

Numerical confidence values remain concealed from candidate-facing feedback views to maintain an encouraging evaluation environment.

TABLE IV
CONTINUOUS SIX-DIMENSIONAL RL STATE SPACE

Index	State Variable	Range	Semantic Definition
0	Performance (P_t)	[0.0, 1.0]	Technical evaluation score S_{final}
1	AvgPerformance ($\bar{P}_t$)	[0.0, 1.0]	Rolling exponential mean of technical scores
2	Confidence (C_t)	[0.0, 1.0]	Composite speech confidence C_{final}
3	Hesitation (H_t)	[0.0, 1.0]	Speech hesitation proxy ($1 - F_{\text{fluency}}$)
4	TimeNorm (T_t)	[0.0, 1.0]	Ratio of elapsed session time to budget
5	Difficulty (D_t)	[0.0, 1.0]	Current question difficulty level / 5

VI. REINFORCEMENT LEARNING FOR ADAPTIVE DIFFICULTY

PrepAired formulates question difficulty selection as a Markov Decision Process (MDP) optimized using Proximal Policy Optimization (PPO).

A. Six-Dimensional State Space ($\mathbf{s}_t$)

The environment state at turn t is represented by normalized continuous vector $\mathbf{s}_t \in [0.0, 1.0]^6$:

$$\mathbf{s}_t = [P_t, \bar{P}_t, C_t, H_t, T_t, D_t]^T, \quad (19)$$

where $P_t = S_{\text{final}}$ represents current turn technical performance, $\bar{P}_t = \text{RollingMean}(P)$ is rolling average technical score, $C_t = C_{\text{final}}$ is communication confidence, $H_t = 1.0 - F_{\text{fluency}}$ represents hesitation proxy, $T_t = t_{\text{elapsed}}/t_{\text{budget}}$ is normalized elapsed session time, and $D_t = D_{\text{current}}/5.0$ represents current difficulty tier normalized across difficulty scale 1–5. Table IV details the state space dimensions.

B. Action Space and Transitions

The agent selects from discrete action space $\mathcal{A} = \{0, 1, 2\}$:

$$a_t = 0 \implies \text{Easier: } D_{t+1} = \max(1, D_t - 1), \quad (20)$$

$$a_t = 1 \implies \text{Same: } D_{t+1} = D_t, \quad (21)$$

$$a_t = 2 \implies \text{Harder: } D_{t+1} = \min(5, D_t + 1). \quad (22)$$

Difficulty level $D_t \in \{1, 2, 3, 4, 5\}$ bounds selected question difficulty.

C. Reward Formulation

The reward function aligns candidate experience within the Zone of Proximal Development, targeting optimal performance equilibrium $S^* = 0.75$:

$$r_t = 1.0 - |S_{\text{final}} - 0.75| - 0.10 \cdot \mathbb{I}(\text{Oscillation}) + 0.05 \cdot C_{\text{final}}, \quad (23)$$

where $|S_{\text{final}} - 0.75|$ penalizes deviation from target challenge level, $\mathbb{I}(\text{Oscillation})$ penalizes repetitive difficulty toggling, and $0.05 \cdot C_{\text{final}}$ provides minor encouragement for confident speech.

VII. GROUNDED LLM FOLLOW-UP AND FEEDBACK

PrepAired decouples the decision to initiate a follow-up probe from natural language generation. Follow-up probing is governed by deterministic evaluator evidence rather than unconstrained LLM judgment.

TABLE V
DETERMINISTIC FOLLOW-UP TRIGGER CONDITIONS

Trigger Name	Evaluation Condition	Pedagogical Objective
Missing Mandatory	Mandatory concept absent	Probe required concept
Detected Misconception	Matched mistake vector	Address specific misunderstanding
Coverage–Reasoning Gap	$S_2 \geq 0.60 \wedge R < 0.35$	Probe logical reasoning failure
Partial-Answer Gap	$0.45 \leq S_{\text{final}} < 0.75$	Probe missing secondary concepts
Depth Probe	$S_{\text{final}} \geq 0.90$	Probe edge cases / trade-offs

A. Deterministic Follow-Up Triggers

A low score alone does not automatically trigger a follow-up. Instead, the orchestrator evaluates structured condition rules summarized in Table V.

B. Grounded Generation and Bounded Sub-Loops

When a trigger condition is satisfied, Qwen2.5-7B-Instruct served through local Ollama generates a targeted follow-up prompt. Prompts are constructed using strict context grounding containing original question text, candidate response, missing concepts, and identified misconceptions. The candidate responds to Follow-up #1, which undergoes technical re-evaluation. If the gap persists, Follow-up #2 is generated with accumulated history. If the gap remains unresolved after two follow-up attempts, the sub-loop terminates to prevent candidate fatigue, generating qualitative feedback and resuming the primary interview workflow.

C. Qualitative Candidate Feedback

Candidate feedback is presented across seven qualitative narrative sections: Your Answer, What You Covered, Important Concepts Missed, What Was Incorrect or Incomplete, How to Improve, Delivery & Communication, and AI Summary. All quantitative scoring metrics ($S_{\text{final}}, S_1, S_2, R, C_{\text{final}}$) are strictly suppressed from the interface view.

VIII. QUESTION SELECTION AND SESSION CONTROL

At session setup, candidates select a primary session length of 5, 15, or 30 questions. The question selection engine manages question retrieval across Easy ($D \in \{1, 2\}$), Medium ($D = 3$), and Hard ($D \in \{4, 5\}$) difficulty pools.

Questions 1 and 2 constitute a deterministic baseline phase. Both questions are sampled independently from the Easy question pool subject to an intra-session non-repetition constraint. PPO-driven adaptive difficulty selection activates starting at Question 3.

Session termination is governed strictly by primary question count. Follow-up questions generated by the LLM state machine are treated as subordinate probes and do not increment the primary question counter. The session terminates when total primary questions asked equals the selected session length.

IX. HYPERPARAMETER TUNING AND PARAMETER SELECTION

System hyperparameter values were selected through empirical iteration on technical answer rubrics and development response sets. Parameter selection focused on balancing precision and recall across concept matching and reasoning detection.

Threshold $\tau_{\text{concept}} = 0.30$ was established by evaluating cosine similarity distributions between candidate sentences and rubric concepts across development scenarios. Setting $\tau_{\text{concept}} > 0.40$ induced false negatives on valid candidate paraphrases, while $\tau_{\text{concept}} < 0.25$ introduced false positives from generic terminology. Reasoning dampening floor $R \leq 0.30$ and dampening factor 0.60 were selected after observing candidate responses with keyword repetition, where structural concept overlap was present without supporting CrossEncoder entailment.

Mistake matching threshold 0.55 and penalty weight 0.07 (capped at 0.30) were tuned to ensure that explicit technical errors deduct appropriate credit without overwhelming base similarity scores. Bonus threshold 0.50 and credit weight 0.03 provide moderate score elevation for advanced insights. Score aggregation weights (0.15 for S_1 , 0.35 for $S_2^{\text{effective}}$, and 0.50 for R) reflect empirical optimization prioritizing CrossEncoder reasoning verification as the primary determinant of technical answer quality.

Follow-up trigger conditions were calibrated across representative candidate performance levels. The partial-answer gap range ($0.45 \leq S_{\text{final}} < 0.75$ with at least one missing concept) targets answers containing useful progress but identifiable gaps, while the coverage–reasoning gap rule ($S_2 \geq 0.60 \wedge R < 0.35$) specifically addresses fluent responses that cite terminology without establishing underlying logical connections.

X. EXPERIMENTAL METHODOLOGY AND VERIFICATION

System reliability was validated using automated software testing, CrossEncoder model evaluation, and an architectural ablation study. Software verification establishes system contract compliance rather than human pedagogical efficacy.

A. Automated Suite Verification

Backend pipeline integrity was validated using Pytest, covering evaluator math, orchestrator state machine transitions, RL environment bounds, and Qwen prompt grounding. Frontend UI rendering and score suppression were verified using Vitest, achieving a 20/20 test pass rate as detailed in Table VI.

B. CrossEncoder Reasoning Model Validation

The CrossEncoder reasoning model (ms-marco-MiniLM-L-6-v2) was validated on domain technical answer pairs. Model evaluation recorded a Spearman rank correlation coefficient of 0.796 against continuous quality reference targets, confirming strong ranking alignment for technical explanation entailment.

TABLE VI
AUTOMATED VERIFICATION SUITE EXECUTION RESULTS

Test Module	Verification Scope	Status
Evaluator Engine	Formula math, caps, dampening logic	Passed
Orchestrator Core	State transitions and follow-up loops	Passed
RL Environment	6D state space and PPO action bounds	Passed
Audio Pipeline	Prosodic feature extraction limits	Passed
Qwen Service	Grounded prompt structure validation	Passed
Vitest Frontend	Score suppression and layout render	20/20 Passed

TABLE VII
IMPLEMENTATION ABLATION: LLM PARTIAL EVALUATION

Configuration	LLM Calls	Avg Latency	RAM Usage	Score Drift
With Partial-Eval	2	73.22 s	14.17 GB	0.00
Without Partial-Eval	1	34.01 s	11.20 GB	0.00

C. LLM Partial-Evaluation Ablation

An architectural ablation evaluated system efficiency with and without a redundant LLM /partial_eval call. Results summarized in Table VII demonstrate that removing redundant LLM evaluation reduced average turn latency from 73.22 s to 34.01 s and RAM pressure from 14.17 GB to 11.20 GB while preserving exact score fidelity (0.00 score drift).

XI. RESULTS AND DISCUSSION

Empirical verification confirms several key system properties. First, the hybrid evaluator effectively prevents keyword stuffing by dampening concept coverage scores when reasoning entailment is low. Second, the PPO difficulty controller smoothly adjusts question selection across difficulty tiers based on the continuous six-dimensional candidate state. Third, deterministic follow-up gating successfully bounds LLM interaction, eliminating hallucinated scoring and unconstrained conversational drift. Fourth, speech confidence separation prevents delivery vocal hesitation from degrading technical correctness evaluations. Finally, the ablation experiment confirms substantial latency and memory optimizations achieved by streamlining LLM calls.

XII. LIMITATIONS AND THREATS TO VALIDITY

Several limitations should be noted. Local CPU execution of Qwen2.5-7B introduces noticeable response latency during follow-up and feedback generation. Speech feature extraction remains sensitive to acoustic background noise and microphone quality. Technical evaluation precision depends directly on the completeness of curated question rubrics. Furthermore, while automated tests confirm system integrity, full validation of candidate learning outcomes requires future large-scale human user studies.

XIII. REPRODUCIBILITY

PrepAired is implemented in Python 3.10 and React 18. The backend utilizes FastAPI, PyTorch 2.0, SentenceTransformers, FAISS-CPU, Stable-Baselines3, and Ollama. All model configurations, evaluation thresholds, and test scripts are preserved within the project codebase.

XIV. CONCLUSION AND FUTURE WORK

This paper presented PrepAired, an adaptive multimodal technical interview preparation framework unifying hybrid answer evaluation, speech confidence modeling, PPO difficulty adaptation, deterministic follow-up gating, and grounded Qwen2.5-7B interaction. The decoupled architecture enforces rigorous scoring standards while delivering personalized practice. Future work includes GPU acceleration for local LLM serving, expansion of technical question rubrics, and formal clinical studies to validate user learning gains.

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